

Set Type

A set is an unordered collection of elements much like a set in mathematics.

The order of elements is not maintained in the sets. It means the elements may not appear in the same order as they are entered into the set.

A set does not accept duplicate elements.

Set is mutable so we can modify it.

Sets are unordered so we can not access its element using index.

Sets are represented using curly brackets { }

a = {10, 20, 30, "GeekyShows", "Raj", 40}

Creating a Set

A set is created by placing all the items (elements) inside curly braces {}, separated by comma. A set does not accept duplicate elements.

Elements can be of different types except mutable element, like list, set or dictionary.

Ex:-

```
a = {10, 20, 30}
```

```
a = {10, 20, 30, "GeekyShows", "Raj", 40}
```

```
a = {10, 20, 30, "GeekyShows", "Raj", 40, 10, 20}
```

Creating Empty Set

We can create an empty set using set() function.

```
a = set()
```

Accessing elements

Sets are unordered so we can not access its element using index.

```
a = {10, 20, "GeekyShows", "Raj", 40}
```

```
a[0]
```

```
print(a)
```

Modifying Elements

Sets are mutable but as we can not access elements using index so we can not modify it

Adding one Element

We can add a new element to set using add() method.

Syntax:-

```
set_name.add(new_element)
```

```
a.add('Python')
```

Adding Multiple Elements

We can add multiple elements to set using update() method.

The update() method can take tuples, lists, strings or other sets as its argument.

Syntax:- set_name.update(elements)

Ex:-

```
a.update([101, 102, 103])
```

Deleting Element

We can delete element using `remove ( )` or `discard( )` methods.

`Remove( )` method raise an error if element doesn't exists while `discard( )` method remains unchanged.

Syntax:- `set_name.remove(element)`

`set_name.discard(element)`

Ex:-

`a.remove('GeekyShows')`

`a.discard('GeekyShows')`

Copying Elements

Copy () Method is used to copy existing set's elements into another set.

Syntax:- `new_set_name = set_name.copy()`

Ex:- `new_a = a.copy()`

Clearing All Elements

Clear () Method is used remove all elements to the set

Syntax:- set_name.clear()

Ex:- a.clear()

Some other Methods of set

- intersection ()
- union ()
- difference ()
- issubset ()
- issuperset ()